

W-O-SC-10 Operator Setup Guide

NNBT/Purpald + guaiacol + ABTS + MALDI — laccase activity assay on the OT-2

This guide is a deck-setup reference for the person loading the robot. It does not replace the run log the protocol prints in the Opentrons app at the start of every run – that log has the **exact** volumes, wells and tube contents for the parameters chosen that day. This document explains the fixed structure that never changes: which slot holds what, which tube each reagent goes in, and the order operator actions happen in.

Before you touch the deck

Open the protocol in the Opentrons app and fill in the runtime parameters (number of enzymes, Bradford concentrations, molar masses, tip start positions, MALDI on/off). **Do not edit the .py file.**

The app's Setup screen will show a coloured, named liquid in every tube, reservoir well and plate well once parameters are set – use it to double-check this guide against your specific run before starting.

1. What this run does

One dilution series feeds three readouts from the same enzymes:

- **NNBT plate** (slot 1, on the Heater-Shaker) – laccase activity via NNBT/Purpald, run in two parallel arms: plain and with guaiacol.
- **ABTS plate** (swapped into slot 5) – a fast kinetic activity readout at A414/A734, read immediately after spiking.
- **MALDI target** (swapped into slot 5 later) – optional, spotted at intervals while the NNBT plate incubates.

The NNBT plate holds up to 8 enzymes plus three negative controls (NC1 = buffer only, NC2 = heat-inactivated enzyme, NC3 = active enzyme with no NNBT in the mix) and a 5-point lactaldehyde standard curve, each in triplicate, in both the plain and guaiacol arms.

2. Deck layout

Slot	Contents	Notes
1	Heater-Shaker + NNBT plate (on the flat-bottom adapter)	Stays on the deck the whole run. North (slot 4) must stay empty.
2, 4, 9, 10, 11	EMPTY	Clearance, not spare space – leave empty.
5	Tube rack → ABTS plate → MALDI target (in that order)	Only one of the three is ever on the deck at a time. The robot pauses and asks you to make each swap.
3	20 µL tip rack	p20 ONLY. Never place a tube rack, reservoir or anything the p300 touches here (see §7).
6	Dilution plate (96-well)	Holds every diluted stock. Swapped for a fresh dilution plate mid-run if MALDI spotting is on.
7	300 µL tip rack	
8	12-well reagent reservoir	North (slot 11) must stay empty.
12	Fixed trash	Do not move or cover.

Load order for slot 5: put the **tube rack** on before starting the run. Keep the ABTS plate and MALDI target staged off-deck (labelled) until the robot pauses and asks for them by name.

3. Tip racks

Two full racks: a 20 µL rack in slot 3 (p20, right mount) and a 300 µL rack in slot 7 (p300, left mount). In the app, set "20 uL tip row/column" and "300 uL tip column" to the first unused tip if you are reusing partial racks – otherwise leave at A1. The app tells you at run start whether the loaded racks have enough tips; if not, keep a spare full rack of each size on hand, because the run will pause mid-way to ask for one.

4. Tube rack (slot 5, step 1 of 3)

A 24-position 1.5 mL tube rack (4 rows A–D × 6 columns). Enzyme stocks fill column-first: enzyme 1 → A1, enzyme 2 → B1, enzyme 3 → C1, enzyme 4 → D1, enzyme 5 → A2, and so on. Two positions are fixed and never move, regardless of how many enzymes are used:

Tube	Contents	Colour
C6 (fixed)	NC2 – heat-inactivated enzyme stock	
D6 (fixed)	Lactaldehyde stock, 100,000 µM (1 M diluted 1:10)	
A5 (fixed)	PaDa-1 neat – ABTS positive control, made up by hand from powder	
B5 (fixed)	PaDa-1 1:1000 – ABTS positive control, made up by hand from powder	

Enzyme positions (A1, B1, C1, D1, A2 ...) hold each enzyme's own stock at whatever concentration your Bradford assay measured – the robot dilutes it to the common target concentration itself. The enzyme chosen as "NC3 uses enzyme #" is drawn from its own tube a second time, so nothing is short-pipetted. **Exact tube volumes needed are in the run log** – they depend on how many enzymes you run and the tip use they need to survive.

5. Reagent reservoir (slot 8)

A 12-well reservoir. All reagent mixes below are **premixed by hand off-deck** before the run – the robot only dispenses them. Pour to the volume shown in the run log (it already includes dead volume); the table below shows which well gets which reagent.

Well(s)	Reagent	Colour
A1, A2	NNBT reaction mix (assay buffer + NNBT)	
A3	Assay / dilution buffer	
A4	Purpald reagent	
A5	Laccase ABTS reaction mix	
A6	PaDa-1 ABTS reaction mix – H ₂ O ₂ instead of Cu ²⁺ , and a different pH. Not interchangeable with the laccase mix.	
A7, A8	Guaiacol reaction mix (buffer + NNBT + guaiacol)	
A9	No-NNBT mix (NC3, plain arm)	
A10	No-NNBT + guaiacol mix (NC3, guaiacol arm)	
A11	milliQ water (MALDI 1:5 dilution) – only if MALDI spotting is on	
A12	MALDI matrix – only if MALDI spotting is on	

Wells shown as two positions (e.g. NNBT mix, guaiacol mix) mean the reagent is split across both if a single 15 mL well would not hold enough for a large run – the run log tells you whether the second well is actually needed.

6. Run sequence & where the robot will pause

Steps 1–5, 7–9 and 12 run automatically. The robot stops and waits for you at every step marked **PAUSE** – read the on-screen message before resuming.

#	Action	Operator note
1	Buffer dispensed into every dilution well	
2	Enzyme, NC3 and heat-inactivated stocks diluted and mixed to target conc.	
3	Lactaldehyde standard curve built by serial dilution	
4	PaDa-1 neat and 1:1000 moved from tubes into dilution wells	rack is about to leave
5	10 µL spikes into all four NNBT plate blocks	
6	PAUSE	Swap the tube rack for the ABTS plate in slot 5.
7	10 µL spikes into the ABTS plate, one sample at a time	
8	Reaction mixes dispensed into the NNBT plate	
9	Both ABTS mixes dispensed by column – always last, since ABTS reacts on contact	column 1 takes the PaDa-1 mix, the rest the laccase mix
10	PAUSE	Take the ABTS plate to the reader immediately (read A414/A734), then seal the NNBT plate. Load the MALDI plate/target if MALDI is on.
11	Incubate 2 h @ 40°C	If MALDI is on, pauses every interval: unseal → robot spots → reseal .
12	PAUSE for Purpald	Remove the plate seal , then resume.
13	Purpald dispensed, developed 10 min, done	Read A530.

7. Three rules that have already cost a run

Rule 1 – Slot 9 can never take a single-nozzle move

The fixed trash in slot 12 sits directly north of slot 9, and the simulator does not model that clearance. Never place labware needing single-tip access in slot 9. The protocol checks this automatically and will refuse to load if violated.

Rule 2 – a HEIGHT is not a DEPTH

A tip positioned above the liquid surface draws air; a tip pressed to a flat well bottom seals against it and moves nothing. Every liquid-handling height in this protocol is computed from the well's remaining volume – if you ever manually jog the robot, do not assume a fixed height works at every fill level.

Rule 3 – the LEFT mount (p300) cannot reach the right-hand deck column

Both pipettes ride the same X carriage ~34 mm apart, against a hard limit at X 418 mm, so the left-mount p300 tops out near deck X 382. Slots 3/6/9 start at x = 265, so anything past x ≈ 117 within one of those slots is physically unreachable by the p300 – and a touch-tip adds another 4 mm on top.

This is what caused the 2026-09-11 run failure: the tube rack was in slot 3 and a touch-tip on tube D6 commanded X422.2, tripping a hard-limit alarm.

Operator rule: slot 3 must only ever hold the 20 µL tip rack. Never place a tube rack, reservoir, or any labware the p300 touches there. The tube rack, ABTS plate and MALDI target all belong in slot 5, which sits in the middle deck column and is reachable by both pipettes.

8. Worked example (3 enzymes) – for illustration only

The tables below are generated from the protocol's own layout code using 3 placeholder enzymes, purely to show the *shape* of the plate maps and what a completed reservoir/tube-rack setup looks like. **Do not pour these volumes for a real run** – use the run log the app prints for your actual parameters.

NNBT plate map (slot 1)

N = NNBT mix · G = guaiacol mix · x = no-NNBT mix (NC3) · y = no-NNBT + guaiacol mix (NC3). Number = group index, keyed to the table below.

	1	2	3	4	5	6	7	8	9	10	11	12
A	N01	N01	N01	G09	G09	G09	N17	N17	N17	G25	G25	G25
B	N02	N02	N02	G10	G10	G10	N18	N18	N18	G26	G26	G26
C	x03	x03	x03	y11	y11	y11	N19	N19	N19	G27	G27	G27
D	N04	N04	N04	G12	G12	G12	N20	N20	N20	G28	G28	G28
E	N05	N05	N05	G13	G13	G13	N21	N21	N21	G29	G29	G29
F	N06	N06	N06	G14	G14	G14	N22	N22	N22	G30	G30	G30
G	N07	N07	N07	G15	G15	G15	N23	N23	N23	G31	G31	G31
H	N08	N08	N08	G16	G16	G16	N24	N24	N24	G32	G32	G32

#	Wells	Content
1	A1,A2,A3	NC1 buffer only
2	B1,B2,B3	NC2 heat-inactivated
3	C1,C2,C3	NC3 (Lac-01)
4	D1,D2,D3	Lac-std 6000 uM
5	E1,E2,E3	Lac-std 4500 uM
6	F1,F2,F3	Lac-std 3000 uM

#	Wells	Content
7	G1,G2,G3	Lac-std 1500 uM
8	H1,H2,H3	Lac-std 500 uM
9	A4,A5,A6	NC1 buffer only
10	B4,B5,B6	NC2 heat-inactivated
11	C4,C5,C6	NC3 (Lac-01)
12	D4,D5,D6	Lac-std 6000 uM
13	E4,E5,E6	Lac-std 4500 uM
14	F4,F5,F6	Lac-std 3000 uM
15	G4,G5,G6	Lac-std 1500 uM
16	H4,H5,H6	Lac-std 500 uM
17	A7,A8,A9	Lac-01
18	B7,B8,B9	Lac-02
19	C7,C8,C9	Lac-03
20	D7,D8,D9	(unused row D)
21	E7,E8,E9	(unused row E)
22	F7,F8,F9	(unused row F)
23	G7,G8,G9	(unused row G)
24	H7,H8,H9	(unused row H)
25	A10,A11,A12	Lac-01
26	B10,B11,B12	Lac-02
27	C10,C11,C12	Lac-03
28	D10,D11,D12	(unused row D)
29	E10,E11,E12	(unused row E)
30	F10,F11,F12	(unused row F)
31	G10,G11,G12	(unused row G)
32	H10,H11,H12	(unused row H)

ABTS plate map (slot 5, swapped in after the dilutions)

#	Wells	Content	Reaction mix
1	A1,B1,C1	PaDa-1 neat	PaDa-1 ABTS mix (H2O2)
2	D1,E1,F1	PaDa-1 1:1000	PaDa-1 ABTS mix (H2O2)
3	A2,A3,A4	NC1 buffer only	laccase ABTS mix
4	B2,B3,B4	NC2 heat-inactivated	laccase ABTS mix
5	C2,C3,C4	Lac-01	laccase ABTS mix
6	D2,D3,D4	Lac-02	laccase ABTS mix
7	E2,E3,E4	Lac-03	laccase ABTS mix

The two PaDa-1 strengths are the positive control – there is no ABTS radical standard curve. They run *down* column 1 because the reaction mixes go in by whole columns, and a column can only take one mix. Wells in a part-filled column still receive reaction mix but hold no sample: **they are blank, do not read them.**

Dilution plate (slot 6), 150 µL per well

Well	From	Stock µL	Buffer µL	Content
A1	res A3	0.0	200.0	NC1 buffer only
B1	tube C6	138.8	61.2	NC2 heat-inactivated
C1	tube A1	138.8	61.2	NC3 (Lac-01)
D1	tube D6	180.0	20.0	Lac-std 6000 uM
E1	tube D6	135.0	65.0	Lac-std 4500 uM
F1	tube D6	90.0	110.0	Lac-std 3000 uM
G1	tube D6	45.0	155.0	Lac-std 1500 uM
H1	tube D6	15.0	185.0	Lac-std 500 uM
A2	tube A1	138.8	61.2	Lac-01
B2	tube B1	180.0	20.0	Lac-02
C2	tube C1	115.7	84.3	Lac-03
D2	res A3	0.0	200.0	(unused row D)
E2	res A3	0.0	200.0	(unused row E)
F2	res A3	0.0	200.0	(unused row F)
G2	res A3	0.0	200.0	(unused row G)
H2	res A3	0.0	200.0	(unused row H)
A3	tube A5	0.0	0.0	PaDa-1 neat
B3	tube B5	0.0	0.0	PaDa-1 1:1000

Reservoir (slot 8) – example pour volumes

Well	Reagent	Volume
A1	NNBT mix	7.8 mL
A3	assay buffer	3.1 mL
A4	Purpald reagent	6.3 mL
A5	laccase ABTS mix	5.6 mL

Well	Reagent	Volume
A6	PaDa-1 ABTS mix (H2O2)	2.5 mL
A7	guaiacol mix	7.8 mL
A9	no-NNBT mix	1.5 mL
A10	no-NNBT guaiacol mix	1.5 mL

Tube rack – example volumes needed

Tube	Contents	Min. volume
A1	NC3 (Lac-01) + Lac-01	≥ 578 µL
B1	Lac-02	≥ 480 µL
C1	Lac-03	≥ 416 µL
A5	PaDa-1 neat	≥ 340 µL
B5	PaDa-1 1:1000	≥ 340 µL
C6	NC2 heat-inactivated	≥ 439 µL
D6	Lac-std 6000 uM + Lac-std 4500 uM + Lac-std 3000 uM + Lac-std 1500 uM + Lac-std 500 uM	≥ 765 µL

Generated from Lac_W-O-SC-10_nnbt_guaiacol_abts_maldi_ot2.py – Kristine Toft Johansen (s215098). This guide reflects the protocol as of the "slot-swap left-mount reach" fix (tube rack confined to slot 5; slot 3 reserved for the 20 µL tip rack only).